Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-05.
Department of Electronics and Telecommunication Engineering

Mini-Project (AY-2024-25)

Synopsis On

Submitted By (Project group members)

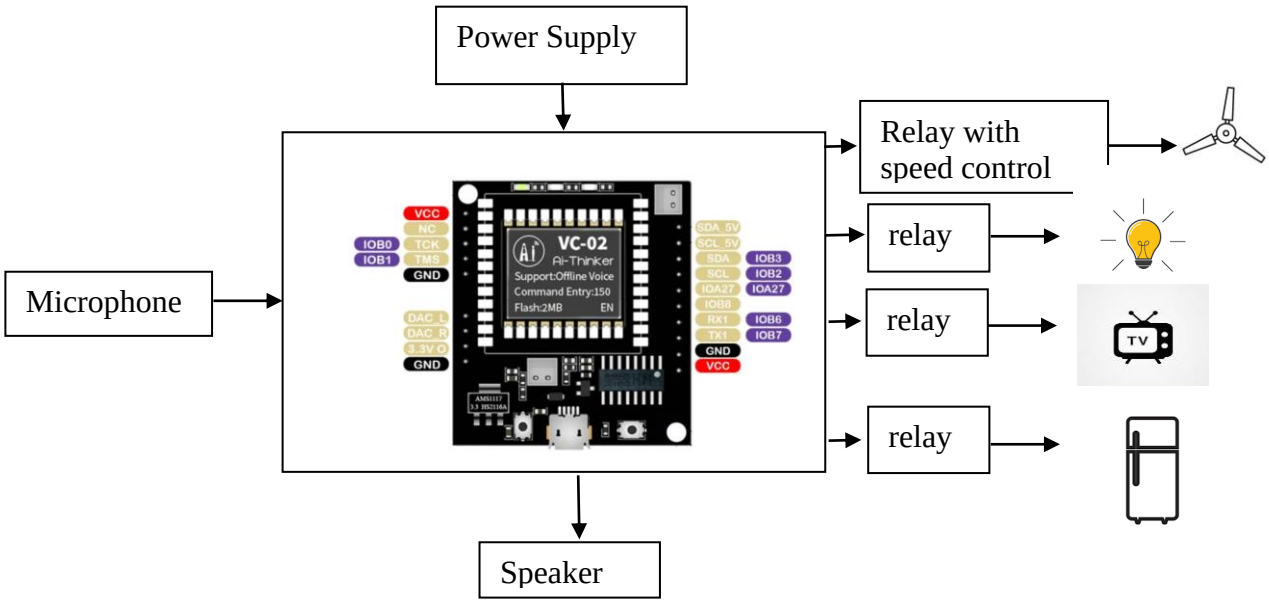
Date:10/1/25

1.Ashitosh Rangnath Jadhav, 32035
2.Sakshi Raju Baravkar ,32007
3.Aditya Milind Bhangre,32010

Div:TE-B

Title of Project 1: Home appliences controlling using voice commands without internet using VC-02 module

Block Diagram:



Introduction of project 1:

This project implements a voice-controlled home automation system. It leverages the VC-02 voice command module, which processes voice commands locally and directly triggers relays to control home appliances. The system is cost-effective, simple to implement, and operates offline, making it an ideal solution for basic smart home automation needs.

The VC-02 directly interfaces with relay modules to switch appliances on or off based on recognized commands, making it a straightforward, efficient, and cost-effective solution for enhancing convenience in home environments.

- **References:**
  http://ieeexplore.ieee.org/xpl/articleDetails.jsp?arnumber=7443538&queryText=voice%20home%20automation&newsearch=true

- **http://ieeexplore.ieee.org/xpl/articleDetails.jsp?arnumber=5937116&queryText=voice%20home%20automation&newsearch=true**

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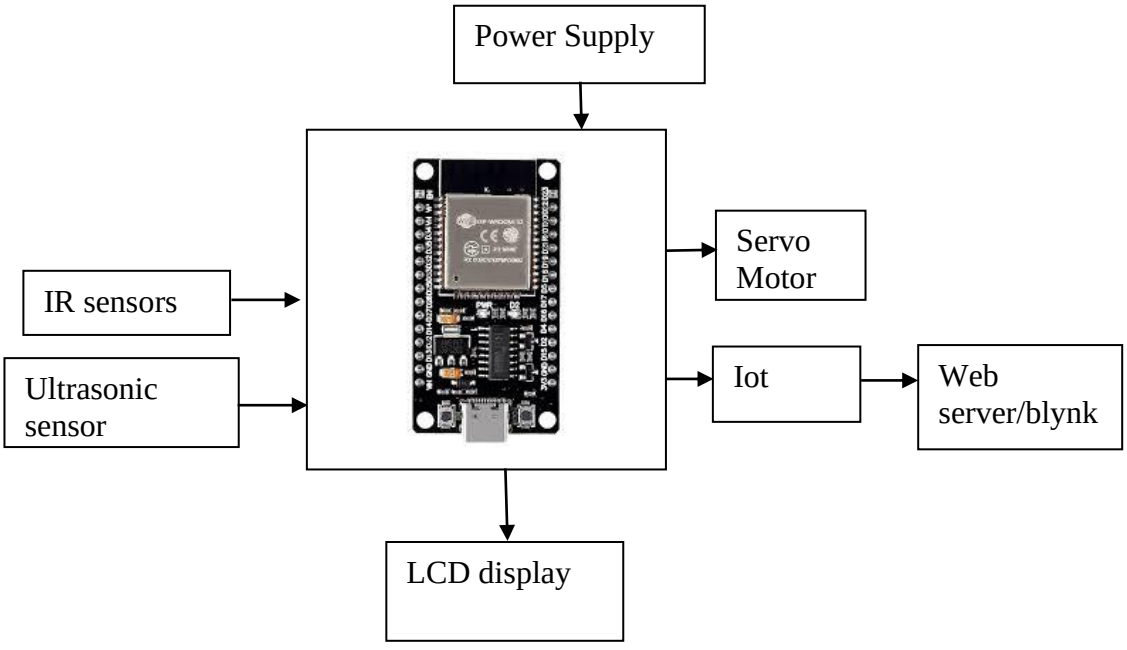
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Title of Project 2: smart car parking system

lock Diagram:



Introduction of project 2:

The Smart Car Parking System is an innovative solution designed to address the challenges of managing parking spaces efficiently. With the growing number of vehicles in urban areas, finding an available parking spot can be time-consuming and frustrating. This system leverages modern technology, including the ESP32 microcontroller, sensors, and IoT connectivity, to streamline the process of parking management.

The system uses ultrasonic sensors to monitor the occupancy of parking slots, providing real-time updates on their availability. The status of each slot is displayed locally on an LCD screen and transmitted to an IoT platform, enabling remote monitoring through a smartphone or web interface. This ensures users can easily identify vacant slots before arriving at the parking lot, saving time and reducing traffic congestion.

By integrating automation and IoT, the Smart Car Parking System offers a scalable, cost-effective, and user-friendly approach to improving parking operations. It is ideal for use in commercial complexes, residential buildings, and public parking spaces, enhancing convenience and contributing to smarter urban infrastructure.

- **References:**

1] **https://www.researchgate.net/publication/345650679_Smart_Car_Parking_System**

2] **https://ieeexplore.ieee.org/document/9087513**

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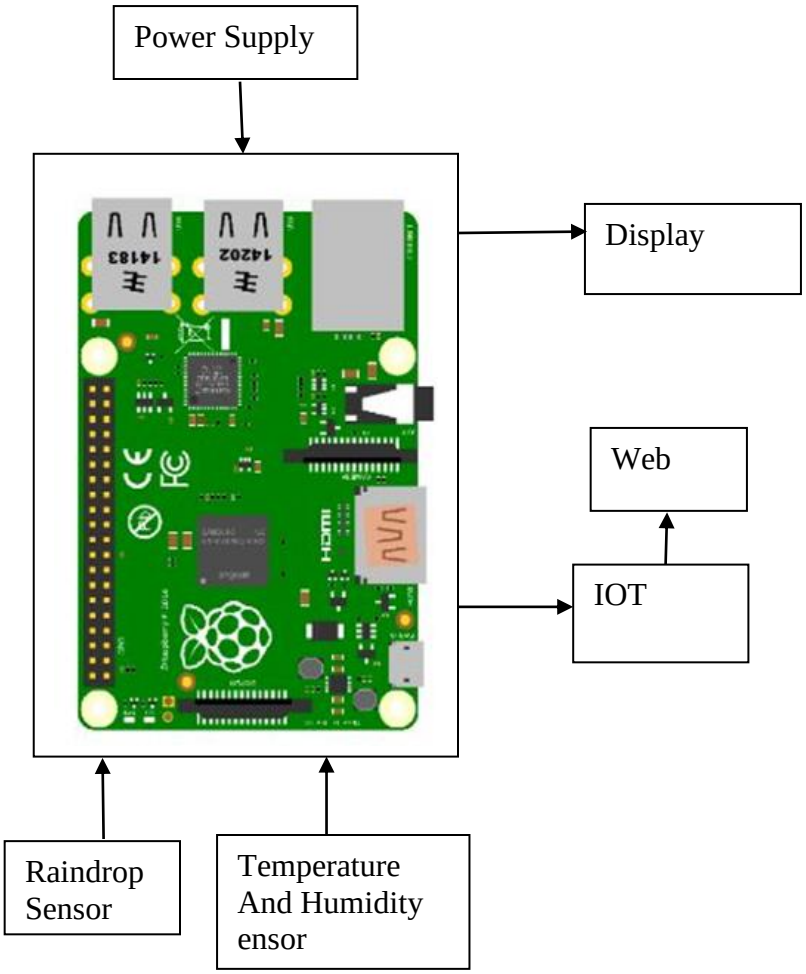
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Title of Project 3: IoT Weather Station using Rassberry-pi

Block Diagram:



Introduction of project 3:

This system can be used to monitor and update weather conditions over the internet. The system monitors 3 parameters namely temperature, humidity and rainfall these values are then displayed on LCD and also updated over the IoT . By readings, the user can get a fair idea of the weather of a particular area on the monitor. This system proves to be useful for knowing the weather of the localized area. The system is powered by Raspberry Pi it includes a raindrop sensor, a temperature sensor, LCD and a buzzer. After turning the system on, the system gets connected to the website by using WIFI. The system keeps track of 3 parameters like temperature, humidity, and rain. As the weather changes the system to monitor and updates the status over the IOT when the area is dry it shows zero value. When the system detects raindrop, it shows the value of the increase in rainfall. When the temperature increases the value gets updated. The user can observe the weather status of a particular area from any remote location.

References:

1]https://www.researchgate.net/publication/332676356_Iot_Based_Weather_Station_Using_Raspberry_Pi_3

2] https://nevonprojects.com/raspberry-pi-based-weather-reporting-over-iot/